

# AI晶片驅動ABF載板供不應求

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- 報告單位：元富投顧研究處
- 報告人：鄭羽涵
- 報告日期：2026年6月12日

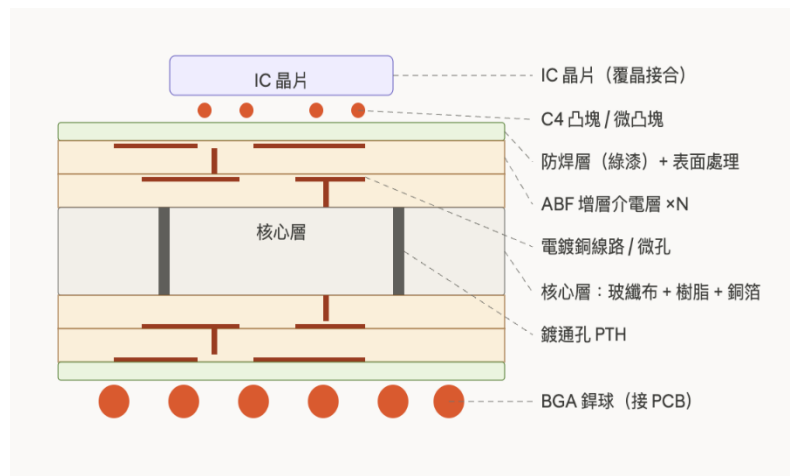
# Agenda

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- ABF載板產業概況
- ABF載板需求
- ABF載板上游材料
  - Ajinomoto、Nittobo、Resonac、MGC
- ABF載板
  - IBIDEN、AT&S、SEMCO
- 個股推薦
  - 欣興(3037)、景碩(3189)
- 結論：ABF載板供需展望

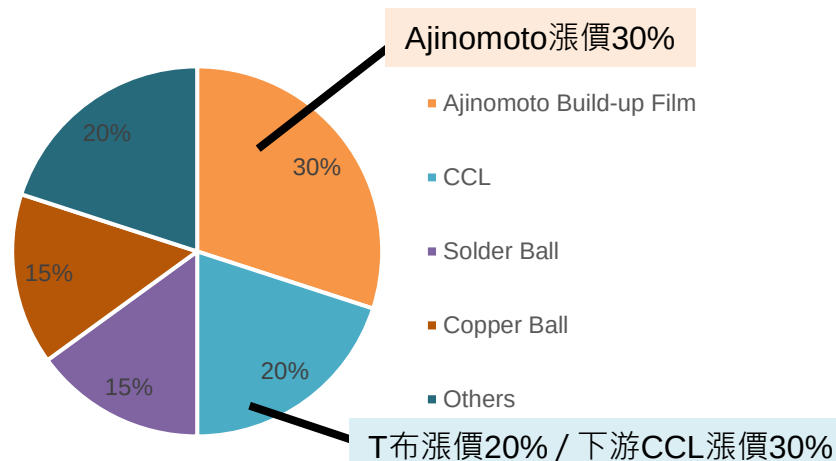
# FC-BGA為高階AI晶片主流封裝載板架構

## ABF載板架構



資料來源：元富投顧

## ABF載板BOM cost

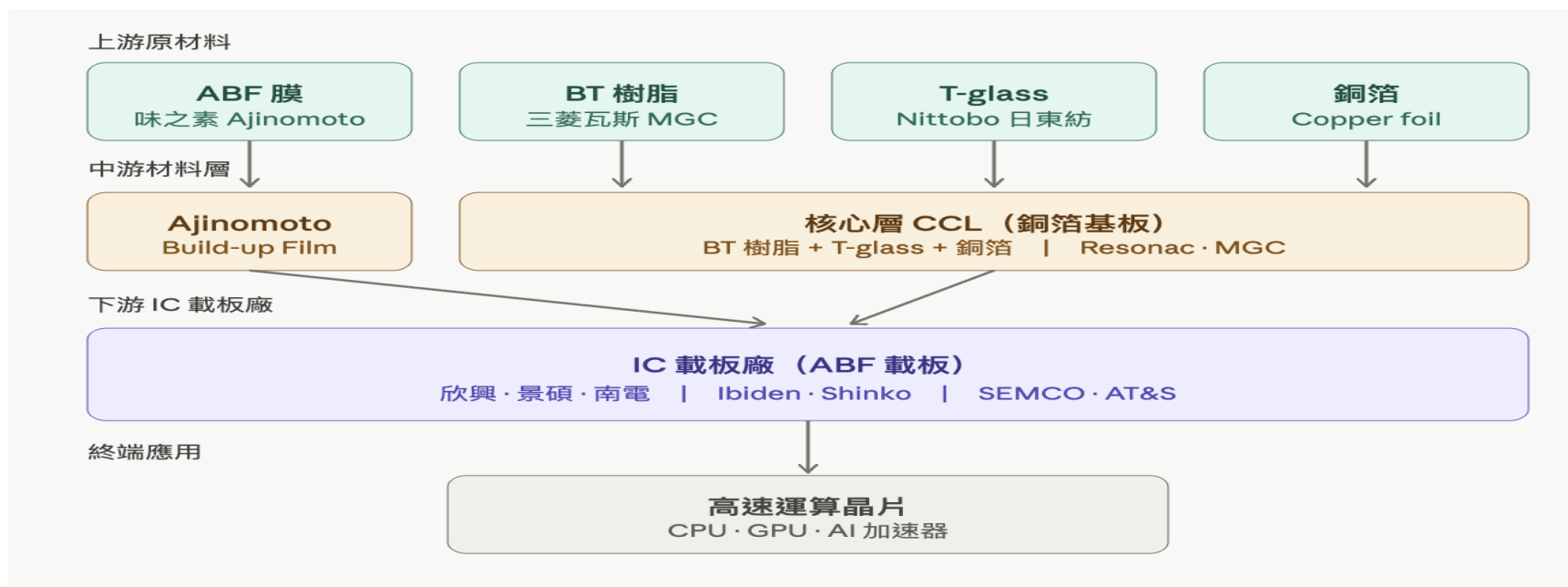


資料來源：元富投顧

|      | ABF載板<br>Ajinomoto Build-up Film | BT載板<br>Bismaleimide Triazine |
|------|----------------------------------|-------------------------------|
| 材料特性 | 低熱膨脹、適合高密度線路                     | 成本較低、成熟穩定                     |
| 技術難度 | 較高                               | 較低                            |
| 線路密度 | 高，可支援細線寬 / 細線距                   | 較低                            |
| 層數   | 較多，適合大尺寸封裝                       | 較少，適合中小尺寸封裝                   |
| 載板尺寸 | 通常較大                             | 通常較小                          |
| 主要應用 | CPU、GPU、AI ASIC、FPGA、HPC、Server  | 手機AP、記憶體、WiFi、RF、MCU、部分消費性IC  |
| 成長驅動 | AI server、HPC、chiplet、大尺寸封裝      | 手機、消費電子、車用與一般 IC              |
| ASP  | 高                                | 低至中                           |

- FC-BGA(Flip Chip Ball Grid Array)為高階CPU、GPU及ASIC最主流封裝載板架構
- 透過晶片翻轉直接與ABF載板連接，大幅提升I/O數量、訊號傳輸速度及供電能力

# ABF載板上下游供應鏈



| ABF載板上游材料                                                | 主要供應商        | 產業地位                              |
|----------------------------------------------------------|--------------|-----------------------------------|
| ABF Film                                                 | Ajinomoto(日) | 全球龍頭，獨佔市場                         |
| T-glass                                                  | Nittobo(日)   | 高階Low-CTE玻布龍頭，AI載板用T-glass市佔率>80% |
| ABF CCL                                                  | Resonac(日)   | 全球ABF載板Core Material龍頭            |
| BT Resin                                                 | MGC(日)       | 全球BT樹脂龍頭，BT載板關鍵材料供應商              |
| 全球ABF高階載板主要廠商                                            |              |                                   |
| ibiden(日)、Shinko(日)、欣興(台)、南電(台)、景碩(台)、AT&S(奧地利)、SEMCO(韓) |              |                                   |

# AI晶片載板走向大尺寸、高層數，推升ABF需求

| Category | Platform          | Vendor    | 2026 Shipment(K)預估 | 2027 Shipment(K)預估 | 載板主要供應商            | 尺寸               | 層數     | 相對PC CPU載板倍數 |
|----------|-------------------|-----------|--------------------|--------------------|--------------------|------------------|--------|--------------|
| GPU      | H200              | NVIDIA    |                    |                    | lbiden / 欣興        | 55x55mm          | 12L    | 4            |
| GPU      | B200              | NVIDIA    |                    |                    | lbiden / 欣興        | 75x85mm          | 16~20L | 12           |
| GPU      | B Ultra           | NVIDIA    | 5,500              |                    | lbiden / 欣興        | 75x85mm          | 16~20L | 12           |
| GPU      | Rubin             | NVIDIA    | 3,000              | 6,000              | lbiden / 欣興        | 95x95~100x100mm  | 18~24L | 21           |
| GPU      | Rubin Ultra       | NVIDIA    |                    | 2,000              | lbiden / 欣興        | 120x120+mm       | 24~30L | 41           |
| GPU      | MI300             | AMD       |                    |                    | 景碩 / 欣興            | 60x60mm          | 12L    | 4            |
| GPU      | MI325             | AMD       |                    |                    | 景碩 / 欣興            | 60x61mm          | 12L    | 4            |
| GPU      | MI350             | AMD       | 400                |                    | 景碩 / 欣興            | 95x95mm          | 22~24L | 21           |
| GPU      | MI450             | AMD       | 400                | 1,000              | 景碩 / 欣興            | 110x110mm        | 26~28L | 33           |
| ASIC     | TPU v5e           | Google    |                    |                    | 欣興 / 南電            | 60x60~70x70mm    | 12L    | 5            |
| ASIC     | TPU v5p           | Google    | 250                |                    | 欣興 / 南電            | 60x60~70x70mm    | 12L    | 5            |
| ASIC     | TPU v6            | Google    | 1,000              |                    | 欣興 / 南電            | 70x70~80x80mm    | 12~16L | 8            |
| ASIC     | TPU v7            | Google    | 1,100              | 2,200              | 欣興 / 南電            | 80x80~90x90mm    | 14~18L | 12           |
| ASIC     | TPU v8Ax          | Google    | 1,600              | 3,100              | 欣興 / 南電            | 90x90~100x100mm  | 18~22L | 18           |
| ASIC     | TPU v8x           | Google    | 600                | 1,800              | 欣興 / 南電            | 90x90~100x100mm  | 18~22L | 18           |
| ASIC     | TPU v8e           | Google    |                    | 300                | 欣興 / 南電            | 90x90~100x100mm  | 18~22L | 18           |
| ASIC     | TPU v8p           | Google    |                    | 500                | 欣興 / 南電            | 90x90~100x100mm  | 18~22L | 18           |
| ASIC     | Trainium 2        | AWS       |                    |                    | 欣興 / 南電            | 80x80mm          | 16~18L | 11           |
| ASIC     | Trainium 2.5      | AWS       | 500                |                    | 欣興 / 南電            | 80x80mm          | 16~18L | 11           |
| ASIC     | Trainium 3        | AWS       | 2,300              | 3,000              | 欣興 / 南電            | 80x80mm          | 16~18L | 11           |
| ASIC     | Trainium 4        | AWS       |                    | 120                | 欣興 / 南電            | 85x85~100x100 mm | 18~24L | 19           |
| ASIC     | Maia M300         | Microsoft | 50                 | 300                | 欣興 / 南電            | 105x105mm        | 24L    | 27           |
| ASIC     | MTIA2             | Meta      | 300                | 200                | 欣興 / 南電            | 85x85mm          | 18L    | 13           |
| ASIC     | MTIA3             | Meta      | 10                 | 300                | 欣興 / 南電            | 100x100mm        | 24L    | 24           |
| CPU      | Sierra Forest     | Intel     |                    |                    | lbiden / Shinko    | 60x60~65x65mm    | 12~14L | 5            |
| CPU      | Clearwater Forest | Intel     |                    |                    | lbiden / Shinko    | 65x65~70x70mm    | 12~16L | 6            |
| CPU      | Granite Rapids    | Intel     |                    |                    | lbiden / Shinko    | 70x70~75x75mm    | 14~18L | 8            |
| CPU      | Diamond Rapids    | Intel     |                    |                    | lbiden / Shinko    | 80x80~90x90mm    | 18~22L | 13           |
| CPU      | Turin             | AMD       |                    |                    | 景碩 / 欣興            | 75x75~80x80mm    | 14~18L | 9            |
| CPU      | Venice            | AMD       |                    |                    | 景碩 / 欣興            | 85x85~95x95mm    | 18~22L | 15           |
| CPU      | Verano            | AMD       |                    |                    | 景碩 / 欣興            | 95x95~100x100mm  | 20~24L | 20           |
| CPU      | Grace             | NVIDIA    | 2750               |                    | 欣興 / 景碩            | 65x55mm          | 12L    | 4            |
| CPU      | Vera              | NVIDIA    | 1500               | 4000               | 景碩(share>50%) / 欣興 | 75x75mm          | 14~16L | 9            |
| CPU      | Graviton          | AWS       |                    |                    |                    | 55x55~65x65mm    | 10~14L | 4            |
| CPU      | Axion             | Google    |                    |                    |                    | 60x60~70x70mm    | 12~16L | 5            |

**CPU:GPU = 1:2**

- Rubin NVL144 : 72 Vera + 144 Rubin
- Rubin Ultra : 144 Vera + 288 Rubin Ultra

# Agentic AI帶動Orchestration CPU需求

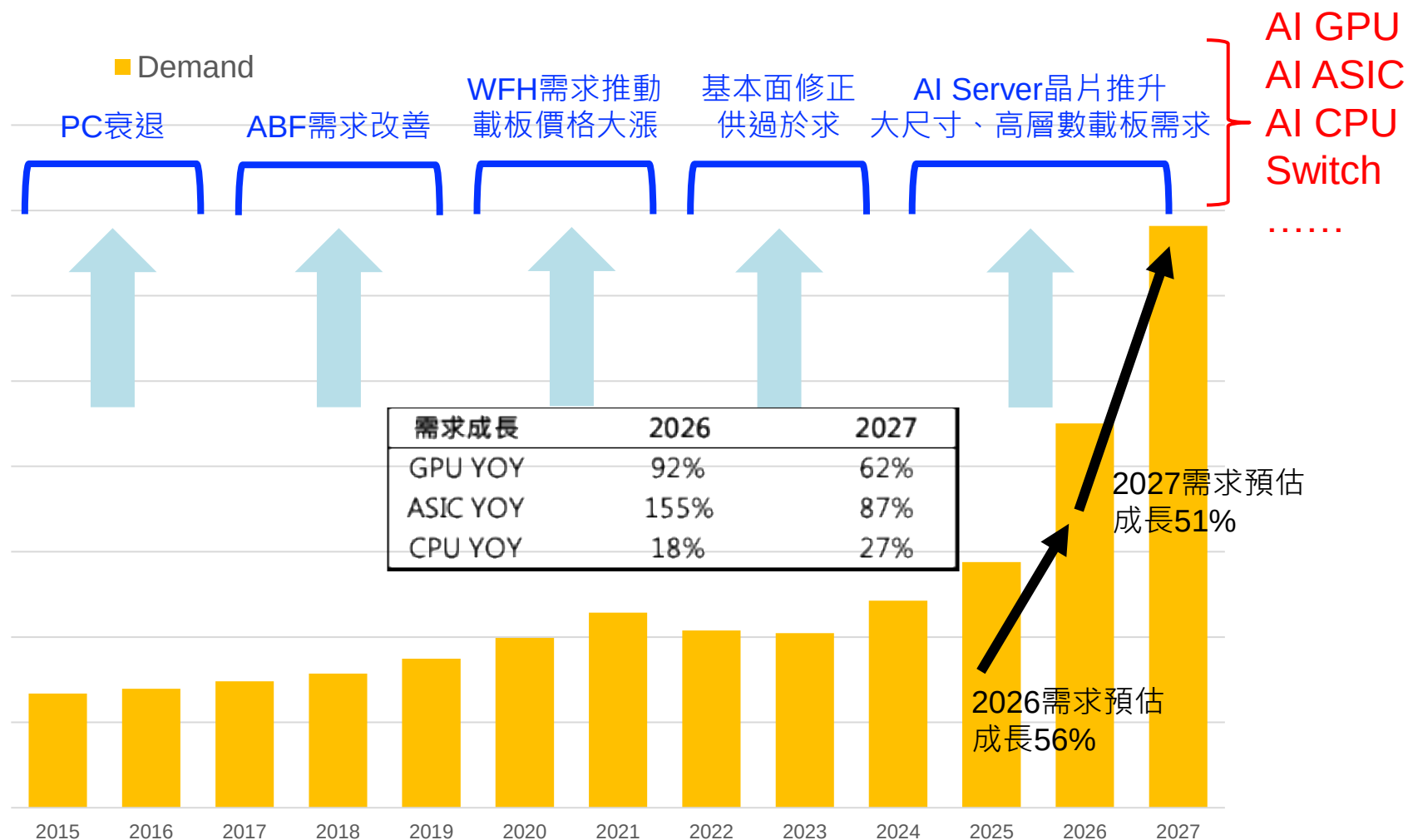
| Category | Platform                    | Vendor | CPU類型                    | 推出時程          | 架構  |
|----------|-----------------------------|--------|--------------------------|---------------|-----|
| CPU      | Sierra Forest (Xeon 6)      | Intel  | Cloud CPU                | 2024推出        | x86 |
| CPU      | Clearwater Forest (Xeon 6+) | Intel  | Cloud CPU                | 2026推出；2027量產 | x86 |
| CPU      | Granite Rapids (Xeon 6)     | Intel  | Host CPU                 | 2024推出        | x86 |
| CPU      | Diamond Rapids (Xeon 7)     | Intel  | Host / Orchestration CPU | 2026推出；2027量產 | x86 |
| CPU      | Turin                       | AMD    | Host CPU                 | 2024推出        | x86 |
| CPU      | Venice                      | AMD    | Host / Orchestration CPU | 2026推出        | x86 |
| CPU      | Verano                      | AMD    | Orchestration CPU        | 2027推出        | x86 |
| CPU      | Grace                       | NVIDIA | Host CPU                 | 2023推出        | ARM |
| CPU      | Vera                        | NVIDIA | Orchestration CPU        | 2026推出        | ARM |
| CPU      | Graviton                    | AWS    | Cloud CPU                |               | ARM |
| CPU      | Axion                       | Google | Cloud CPU                |               | ARM |



資料來源：TrendForce、元富投顧

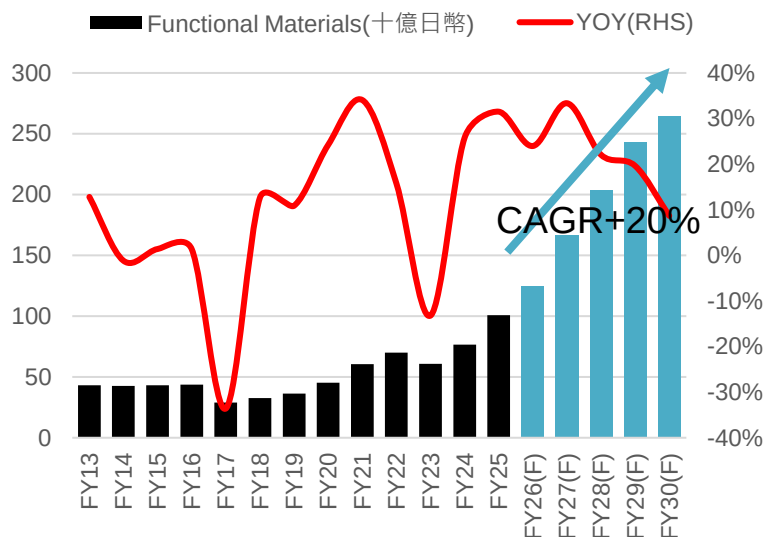
|                      | 2025 | 2026F | 2027F | 2028F | 2029F | 2030F |
|----------------------|------|-------|-------|-------|-------|-------|
| Server CPU Units (M) | 26   | 30    | 32    | 34    | 36    | 38    |
| YOY                  |      | 15%   | 7%    | 6%    | 5%    | 5%    |
| Host                 | 50%  | 48%   | 46%   | 44%   | 42%   | 40%   |
| Cloud                | 50%  | 48%   | 46%   | 44%   | 42%   | 40%   |
| Orchestration        | 0%   | 4%    | 8%    | 12%   | 16%   | 20%   |
| Host                 | 13.0 | 14.4  | 14.8  | 15.0  | 15.0  | 15.0  |
| Cloud                | 13.0 | 14.4  | 14.8  | 15.0  | 15.0  | 15.0  |
| Orchestration        | 0.0  | 1.2   | 2.6   | 4.1   | 5.7   | 7.5   |
| 相對PC CPU載板倍數         |      |       |       |       |       |       |
| Host                 | 8    | 8     | 10    | 10    | 10    | 10    |
| Cloud                | 5    | 5     | 5     | 5     | 5     | 5     |
| Orchestration        | 10   | 10    | 12    | 12    | 15    | 15    |

# AI GPU ASIC CPU推動載板需求暴增



# 上游材料：Ajinomoto展望樂觀且宣布擴廠

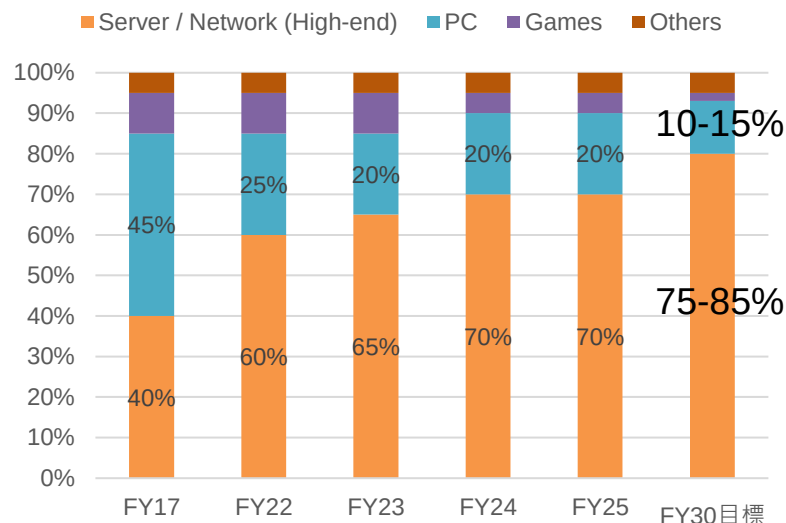
Functional Materials(Build-up film)營收



資料來源：Ajinomoto、Bloomberg、元富投顧

| 時間   | 產能與擴產計劃     |
|------|-------------|
| FY23 | 宣布250億日圓擴產  |
| FY25 | 群馬新廠投產      |
| FY26 | 宣布投資第三ABF基地 |
| FY32 | 第三ABF基地投產   |

ABF終端應用逐年變化



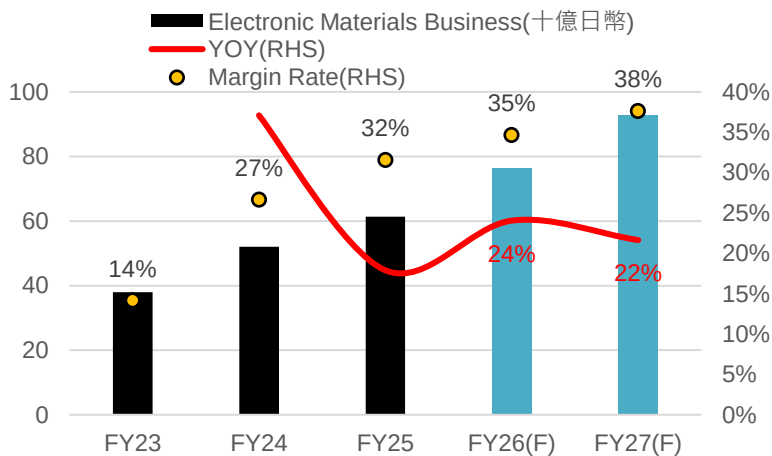
資料來源：Ajinomoto、元富投顧

- FY18 Functional Materials事業獲利率僅30%，FY25獲利率提升至>50%
- 調漲ABF 30%，預計3Q26開始生效
- 2026/5月宣布取得岐阜縣土地，規劃興建第三座ABF生產基地，預計FY32投產，為因應2030年後持續擴大的高階封裝需求



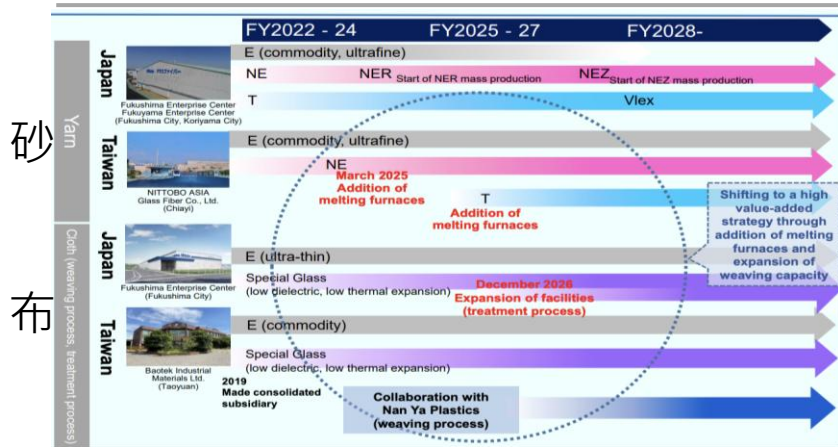
# 上游材料：Nittobo T-glass將持續緊俏至2027年

Electronic Materials(玻砂、玻布)營收與獲利率 玻纖布種類與終端應用



資料來源：Nittobo、Bloomberg、元富投顧

## Nittobo擴廠計畫



資料來源：Nittobo、元富投顧

| Type                                          | E-glass   | T-glass (Low CTE) | NE-glass (Low Dk)               | NER-glass (Low Dk2)             | NEZ-glass                       | Q-glass |
|-----------------------------------------------|-----------|-------------------|---------------------------------|---------------------------------|---------------------------------|---------|
| Dk @ 10 GHz                                   | 6.6       | 5.3               | 4.7                             | 4.5                             | 4.0                             | 3.7     |
| Df @ 1 GHz                                    | 0.006     | 0.007             | 0.0025                          | 0.0018                          | 0.001                           | 0.0005  |
| CTE (ppm/°C)                                  | 5.6       | 2.8               | 3.3                             | 3.4                             | 3.1                             | 0.5     |
| Tensile Modulus                               | 75 GPa    | 86 GPa            | 64 GPa                          | 50 GPa                          | 46 GPa                          | 78 GPa  |
| Composition (SiO <sub>2</sub> )               | 52-56%    | 64-66%            | 45-55%                          | 45-55%                          | 50-55%                          | 99.9%   |
| Composition (Al <sub>2</sub> O <sub>3</sub> ) | 12-16%    | 24-26%            | 10-15%                          | 10-18%                          | 10-18%                          | -       |
| CCL Application                               | M4-M6     | -                 | M7-M8                           | -                               | M9-M10                          | -       |
| End Application                               | Commodity | IC Substrate      | AI Server, 400G switch, HDU/PTH | AI Server, 800G switch, HDU/PTH | AI Server, 1.6T switch, HDU/PTH | -       |

| Substrate type                  |              | Required performance   | Glass fiber type          |                    |
|---------------------------------|--------------|------------------------|---------------------------|--------------------|
|                                 |              |                        | High-end                  | Middle-end         |
| Semiconductor package substrate | CPU/GPU/ASIC | Low CTE                | T                         | E                  |
|                                 | LPUI/DPU     | Low CTE                | T                         |                    |
|                                 | NAND memory  | Low CTE                | T                         | E                  |
|                                 | DDR5 memory  | Low CTE                | T                         |                    |
|                                 | LPDDR memory | Low CTE                | T                         |                    |
|                                 | DDR memory   | Low dielectric tangent | NE / NER                  | E                  |
| Servers(AI/General)             |              | Low dielectric tangent | NE / NER                  | E                  |
| Switch(Core,AI)                 |              | Low dielectric tangent | NE / NER / NEZ            | E                  |
| Expansion cards (LPUI/DPU/NIC)  |              | Low dielectric tangent | NE / NER                  | E                  |
| Semiconductor package substrate | AP-CPU/INPU  | Low CTE                | Ultra-thin T, T           | Ultra-thin E       |
|                                 | NAND memory  | Low CTE                | Ultra-thin T              | Super ultra-thin E |
|                                 | DDR memory   | Low CTE                | Ultra-thin T (Smartphone) |                    |
|                                 | DDR memory   | Low dielectric tangent | NE(PC)                    |                    |
| Motherboard                     |              | Low dielectric tangent | Ultra-thin NE             | Ultra-thin E       |
| RF package substrate            |              | Low dielectric tangent | Ultra-thin NE             | Ultra-thin E       |
| Semiconductor package substrate | CPU/GPU/INPU | Low CTE                | T                         | E                  |
|                                 | DDR memory   | Low dielectric tangent | NE(PC)                    | E                  |
| Semiconductor package substrate |              | Low CTE                | T                         | Ultra-thin E       |
| Semiconductor package substrate |              | Low CTE                | T                         | E                  |
| Module board                    |              | Low dielectric tangent | Ultra-thin NE             | E                  |

NER(Low DK2)

800G、1.6T交換器

NEZ

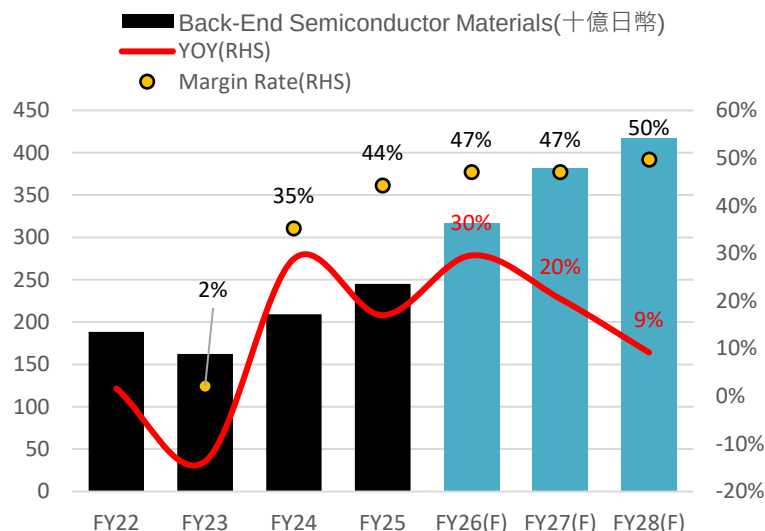
3.2T交換器、CPO

資料來源：TrendForce、Nittobo、元富投顧

- 2025/8月宣布T-glass漲價20%；2026/5月意外宣布不續漲，但CAPEX自800億日圓上調至1200億日圓，主要用於T-glass擴產，目標至2028年產能擴充3倍
- 擴產規劃與時程：2025/3月於嘉義廠新增熔爐(NE、T-glass)、2026/12月福島廠擴充後段處理；2027年產能逐步爬坡，預期2H27 T-glass才會稍微緩解
- 新供應商認證：台廠台玻、富喬

# 上游材料：Resonac MGC 2026年初相繼喊漲

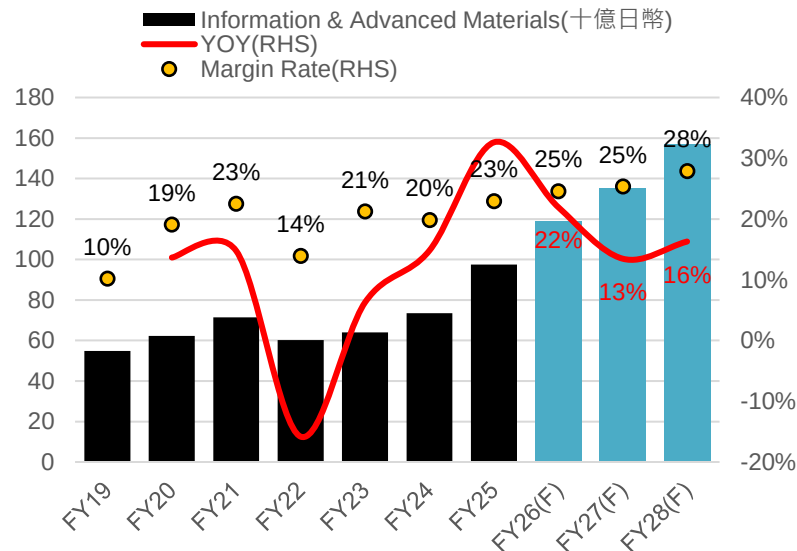
## Resonac Back-end Semi Materials營收



資料來源：Resonac、Bloomberg、元富投顧

- Resonac Back-end Semiconductor Materials產品涵蓋ABF CCL、NCF、TIM等，FY1Q26營收創歷史新高，YOY+38%
- 2026/1月宣布高階ABF CCL漲價30%，FY1Q26成長主要來自出貨量增加，漲價效益尚未完全反映，價格協商持續進行中
- Semiconductor & Electronic Materials展望大幅上修，係因AI材料需求優於預期

## MGC Information & Advanced Materials營收



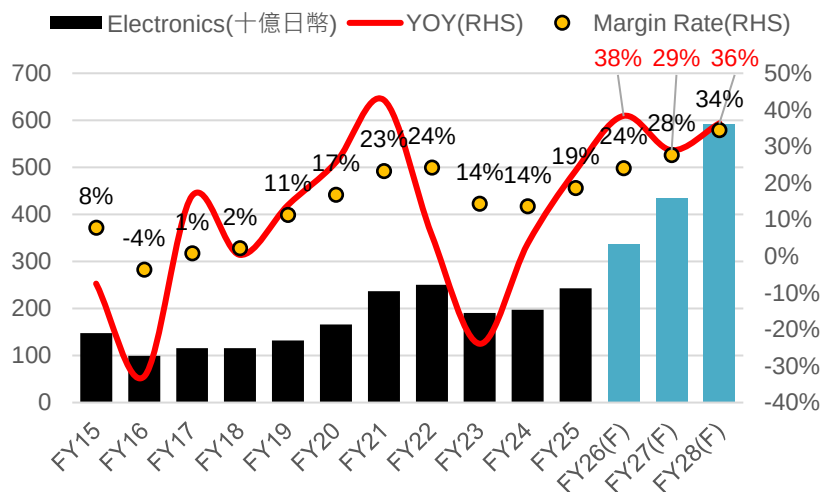
資料來源：MGC、Bloomberg、元富投顧

- MGC-BT樹脂龍頭：泰國BT材料廠擴產完成，FY26開始貢獻營收成長
- 2026年宣布漲價：受上游原料漲價影響，CCL / Prepreg / CRS漲價30%，4月起生效

註：MGC Information & Advanced Materials現改為ICT Business

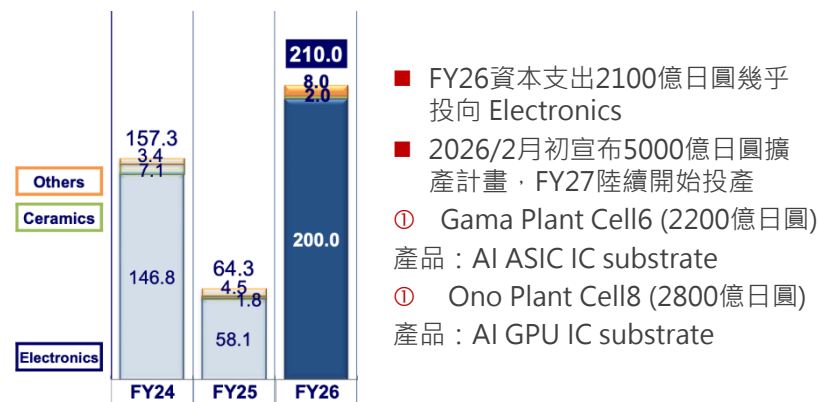
# ABF載板：龍頭IBIDEN上修財測並投資擴產

## IBIDEN Electronics(IC Substrate)營收



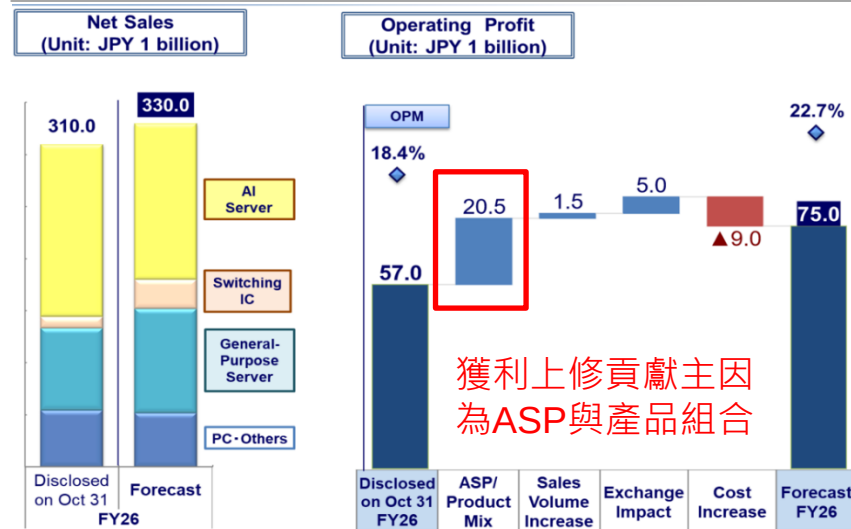
資料來源：IBIDEN、Bloomberg、元富投顧

## IBIDEN資本支出與5000億日圓擴產計畫



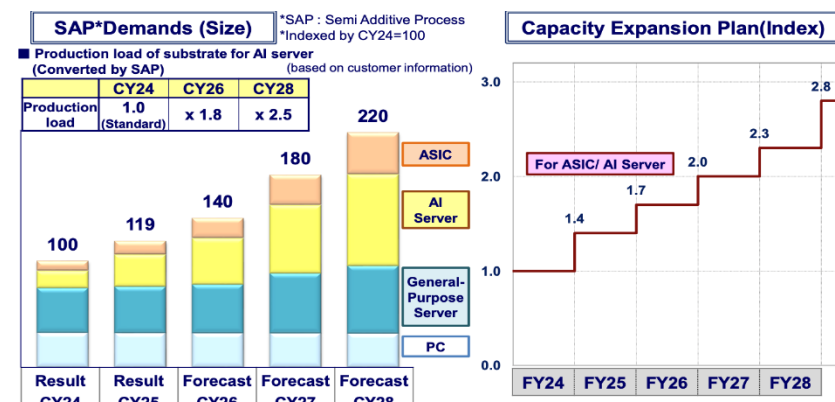
資料來源：IBIDEN、Bloomberg、元富投顧

## IBIDEN上修FY26 Electronics營收、獲利



資料來源：IBIDEN、Bloomberg、元富投顧

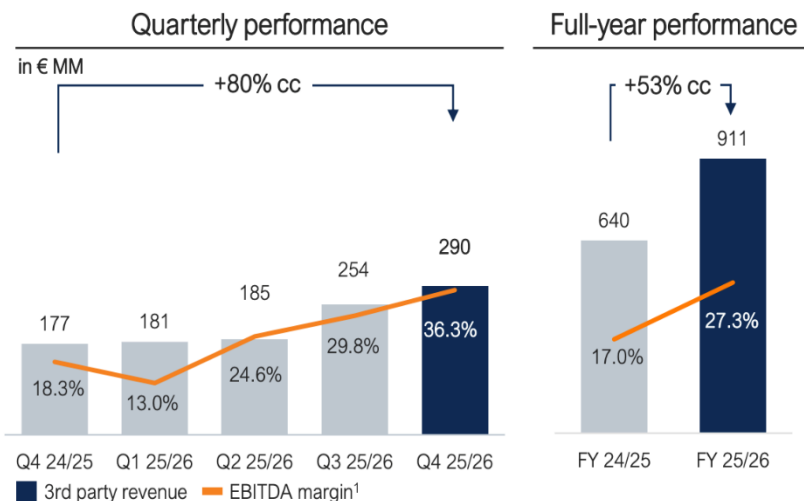
## IBIDEN：載板將供不應求



資料來源：IBIDEN、Bloomberg、元富投顧

# ABF載板：AT&S SEMCO皆提及客戶推動擴產

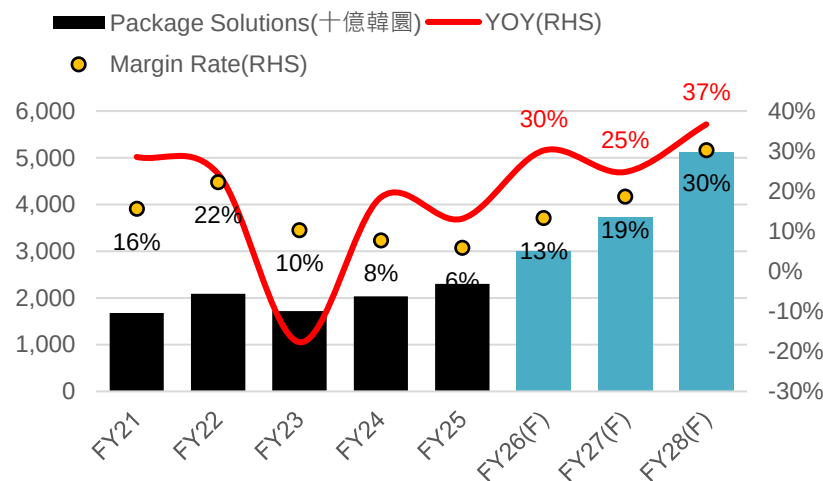
## AT&S Microelectronics(ABF載板)營收



資料來源：AT&S、元富投顧

- 1Q26已調漲ABF載板價格；ABF產業已從稼動率改善 → 進入到價格上漲階段
- 6/15日宣布在馬來西亞投資20億歐元擴產，全面擴建Kulim生產基地，並啟用原本閒置的Kulim二期工廠
- AT&S提及奪下AMD與另一家科技巨頭訂單；第二家客戶身分市場認為可能為Intel
- 目前掌握至少五家美國科技巨頭的長期訂單，因此有足夠資金擴大產能

## SEMCO Package Solutions營收

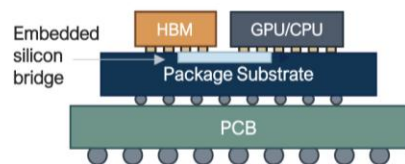


資料來源：SEMCO、Bloomberg、元富投顧

- SEMCO 1Q26 FCBGA營收YOY+45%，為三大事業群中成長最快；AI Accelerator、Server CPU及Networking相關高階載板出貨持續增加，公司預期2Q26高階FCBGA需求仍將維持強勁
- 公司聚焦高階FCBGA產能與大尺寸、高層數、Embedded載板產品，亦表示若新客戶正式放量，後續通常也會推動擴產需求

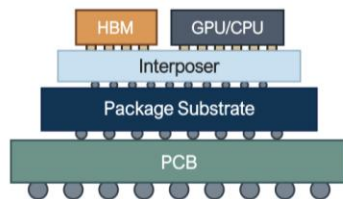
# EMIB推動ABF載板技術升級

**EMIB(-T)**  
Embedded Multi-die Interconnect Bridge  
(with Through-Silicon Vias)



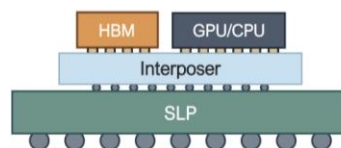
- Cutting-edge substrate technology
- Moving complexity to the substrate
- Scales cost efficient with size

**CoWoS / CoPoS**  
Chip-on-Wafer-on-Substrates /  
Chip-on-Panel-on-Substrate



- Advanced substrate technology
- Complexity balanced between interposer & substrate
- Size scaling requires larger interposer

**CoWoP**  
Chip-on-Wafer-on-PCB



- Essentially merging Substrate and PCB technology
- Several new technology to be industrialized
- Size scaling requires larger interposer

| 項目             | EMIB | EMIB-T(TSV) |
|----------------|------|-------------|
| 互連方式           | 2.5D | 2.5D+3D     |
| Silicon Bridge | V    | V           |
| TSV整合          | X    | V           |
| 頻寬             | 高    | 更高          |
| 功耗             | 較低   | 更低          |
| AI GPU適用性      | 高    | 非常高         |
| HBM擴充能力        | 中    | 高           |
| ABF需求          | 高    | 更高          |

| 項目     | EMIB            | CoWoS              |
|--------|-----------------|--------------------|
| 技術架構   | Silicon Bridge  | Silicon Interposer |
| 訊號傳輸主體 | ABF+Bridge      | Interposer         |
| ABF重要性 | 高               | 中                  |
| ABF面積  | 非常大             | 大                  |
| ABF層數  | 非常高             | 高                  |
| 成本     | 較低              | 高                  |
| 封裝尺寸擴展 | 較容易             | 受Interposer限制      |
| 適用產品   | CPU、ASIC、AI GPU | AI GPU             |

- IBIDEN、欣興為Intel EMIB載板供應商
- EMIB將矽橋嵌入載板內部，實現Chiplet高速互連，可視為Embedded技術早期應用

- **Embedded Die**：進一步將主動晶片嵌入載板結構中，相較EMIB僅嵌入Bridge，可提供更短訊號路徑、更低功耗及更佳系統整合能力

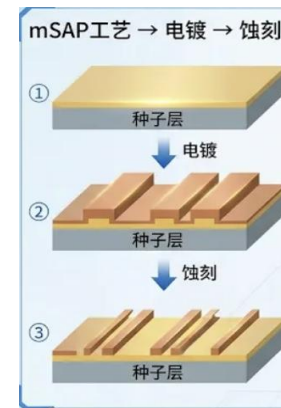
## ■ 玻璃基板(Glass Core)

- ① 隨AI晶片尺寸與HBM數量持續增加，傳統ABF載板面臨翹曲與尺寸限制，推動Glass Core技術發展
- ② Glass Core具備低熱膨脹係數(CTE)、高平坦度及大尺寸製造優勢，可支援超大型Chiplet架構與高密度HBM整合



# mSAP成為1.6T以上光模組關鍵技術

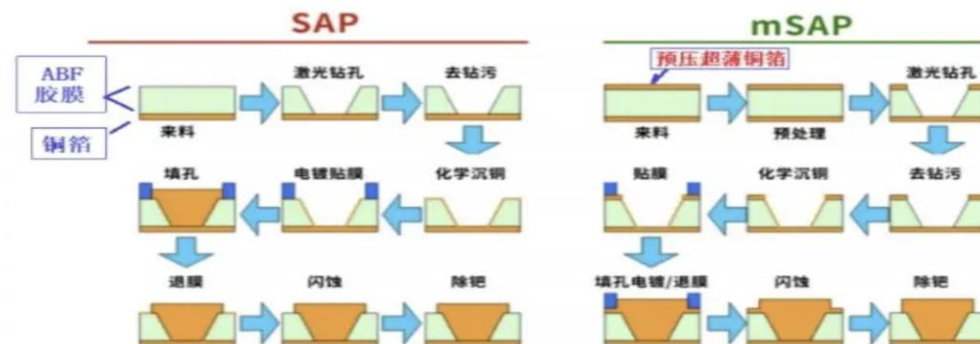
|             | 傳統減成法 (Subtractive Process) | SAP / mSAP (Semi-Additive Process) |
|-------------|-----------------------------|------------------------------------|
| 製程原理        | 從整片銅箔開始，透過蝕刻去除多餘銅層形成線路      | 先形成種子層，再透過電鍍增厚銅線形成線路               |
| 線路形成方式      | 去銅 (Etching)                | 長銅 (Plating)                       |
| 線路截面        | 梯形 (Trapezoid)              | 接近矩形 (Rectangular)                 |
| 線寬控制能力      | 較差，易受蝕刻影響                   | 較佳，尺寸精度高                           |
| 線寬/線距 (L/S) | 25-50 $\mu\text{m}$         | 5-15 $\mu\text{m}$                 |
| 配線密度        | 中                           | 高                                  |
| 訊號完整性       | 較差                          | 較佳                                 |
| 高頻損耗        | 較高                          | 較低                                 |
| 製造成本        | 較低                          | 較高                                 |
| 主要應用        | 一般PCB、HDI板                  | SLP、AI光模組、高速交換器、AI伺服器PCB           |



| 光模塊世代 | 通道速率       | PCB層數  | mSAP   | CCL材料 |
|-------|------------|--------|--------|-------|
| 400G  | 56G PAM4   | 8-10層  | 部分導入   | M4/M5 |
| 800G  | 112G PAM4  | 12-14層 | 主流     | M6    |
| 1.6T  | 224G PAM4  | 14-16層 | 必要     | M7    |
| 3.2T  | 224G+ PAM4 | 16-20層 | 高階mSAP | M8+   |

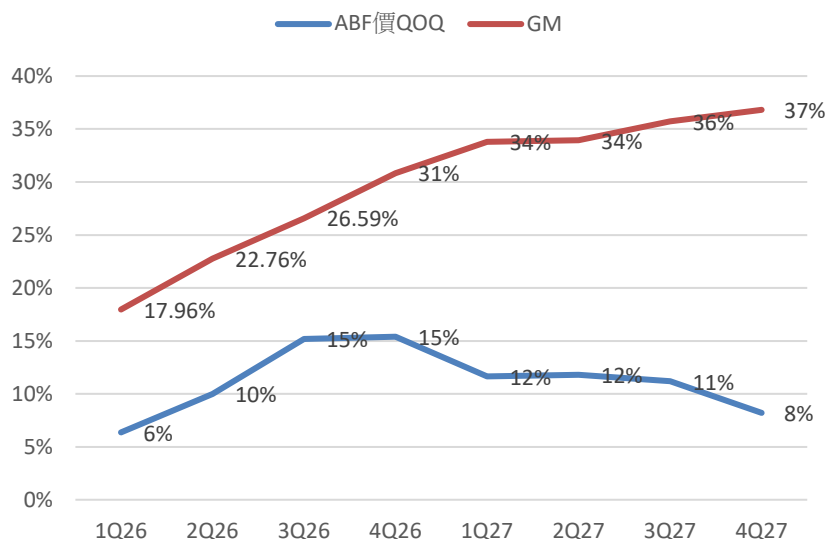
| 製程       | 線寬/線距(L/S)          |
|----------|---------------------|
| 傳統HDI    | 30-50 $\mu\text{m}$ |
| mSAP SLP | 10-15 $\mu\text{m}$ |
| 高階SLP    | 8-10 $\mu\text{m}$  |
| BT載板     | 5-10 $\mu\text{m}$  |
| ABF載板    | 2~5 $\mu\text{m}$   |

- 為什麼光模組需要mSAP？**高速傳輸帶動細線路需求**：mSAP可實現10-15 $\mu\text{m}$ 以下線寬/線距，遠優於傳統減成法30 $\mu\text{m}$ 以上的極限；且mSAP線路截面接近矩形，有助於提升訊號品質

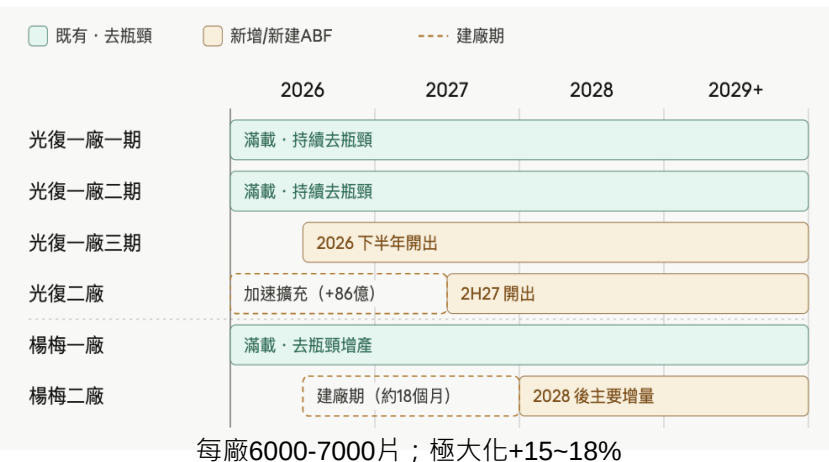


# 欣興(3037 TT,TP=NT\$1500)載板龍頭廠產能爆滿

| Sales(NT\$m) | 2025    | %    | YoY  | 2026(F) | %    | YoY | 2027(F) | %    | YoY |
|--------------|---------|------|------|---------|------|-----|---------|------|-----|
| Substrate    | 76,127  | 58%  | 7%   | 114,188 | 62%  | 50% | 211,874 | 71%  | 86% |
| ABF          | 60,093  | 46%  | 19%  | 98,311  | 54%  | 64% | 195,997 | 66%  | 99% |
| BT           | 16,034  | 12%  | -22% | 15,877  | 9%   | -1% | 15,877  | 5%   | 0%  |
| HDI          | 35,736  | 27%  | 28%  | 45,176  | 25%  | 26% | 56,470  | 19%  | 25% |
| PCB          | 14,815  | 11%  | 23%  | 19,646  | 11%  | 33% | 24,558  | 8%   | 25% |
| FPC          | 3,250   | 2%   | -6%  | 3,097   | 2%   | -5% | 3,097   | 1%   | 0%  |
| Other        | 1,312   | 1%   | 14%  | 1,639   | 1%   | 25% | 1,885   | 1%   | 15% |
| 合計           | 131,241 | 100% | 14%  | 183,746 | 100% | 40% | 297,883 | 100% | 62% |



## 欣興ABF高階廠區擴產計畫：



# 欣興(3037 TT,TP=NT\$1500)載板龍頭廠產能爆滿

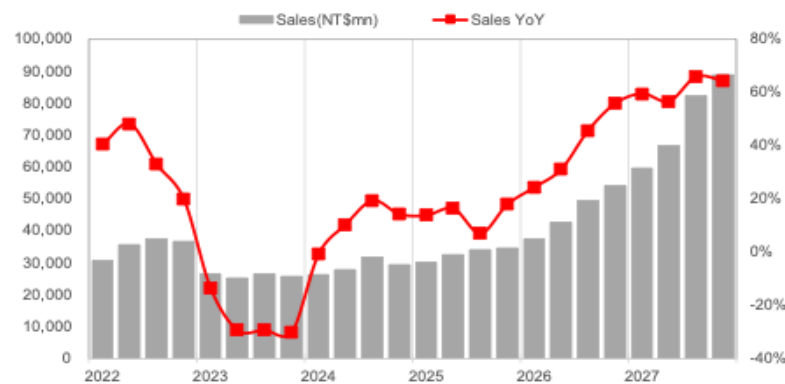
## Comprehensive income statement

NT\$m

| Comprehensive income statement |              |              |               |               | NT\$m |
|--------------------------------|--------------|--------------|---------------|---------------|-------|
| Year-end Dec. 31               | FY24         | FY25         | FY26          | FY27 F        |       |
|                                | IFRS         | IFRS         | IFRS          | IFRS          |       |
| Net sales                      | 115,373      | 131,241      | 183,746       | 297,883       |       |
| COGS                           | 99,058       | 112,949      | 137,453       | 192,846       |       |
| Gross profit                   | 16,315       | 18,292       | 46,293        | 105,037       |       |
| Operating expense              | 11,265       | 11,782       | 17,243        | 27,959        |       |
| <b>Operating profit</b>        | <b>5,117</b> | <b>6,674</b> | <b>29,075</b> | <b>77,078</b> |       |
| Total non-operate. Inc.        | 2,204        | 2,153        | 13,367        | 1,862         |       |
| Pre-tax profit                 | 7,321        | 8,827        | 42,442        | 78,940        |       |
| Total Net profit               | 5,556        | 7,550        | 32,605        | 59,375        |       |
| Minority                       | 0            | 0            | 0             | 0             |       |
| <b>Net Profit</b>              | <b>5,082</b> | <b>6,673</b> | <b>30,558</b> | <b>55,648</b> |       |
| <b>EPS (NT\$)</b>              | <b>3.35</b>  | <b>4.38</b>  | <b>19.34</b>  | <b>35.02</b>  |       |
| Y/Y %                          | FY24         | FY25         | FY26          | FY27 F        |       |
| Sales                          | 10.9         | 13.8         | 40.0          | 62.1          |       |
| Gross profit                   | 2,250.0      | 12.1         | 153.1         | 126.9         |       |
| Operating profit               | 2,028.7      | 30.4         | 335.6         | 165.1         |       |
| Pre-tax profit                 | 1,841.8      | 20.6         | 380.8         | 86.0          |       |
| Net profit                     | 5,865.2      | 31.3         | 357.9         | 82.1          |       |
| EPS                            | -14.3        | 30.7         | 341.4         | 81.1          |       |
| Margins %                      | FY24         | FY25         | FY26          | FY27 F        |       |
| Gross                          | 14.1         | 13.9         | 25.2          | 35.3          |       |
| Operating                      | 4.4          | 5.1          | 15.8          | 25.9          |       |
| EBITDA                         | 7.0          | 7.5          | 0.0           | 0.0           |       |
| Pre-tax                        | 6.3          | 6.7          | 23.1          | 26.5          |       |
| <b>Net</b>                     | <b>4.4</b>   | <b>5.1</b>   | <b>16.6</b>   | <b>18.7</b>   |       |

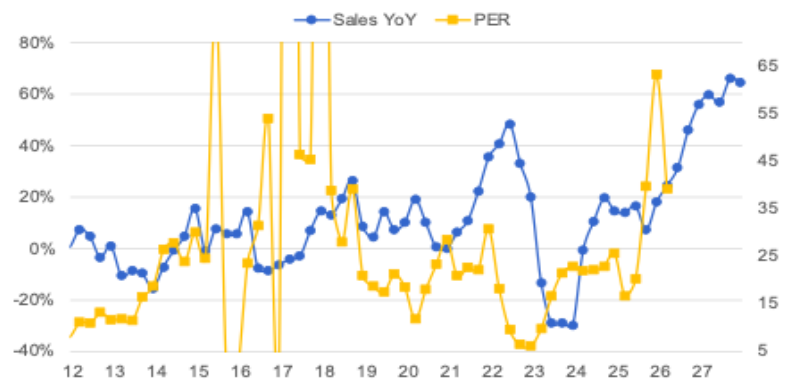
資料來源：元富投顧

## 營收與營收YOY



資料來源：元富投顧

## PER

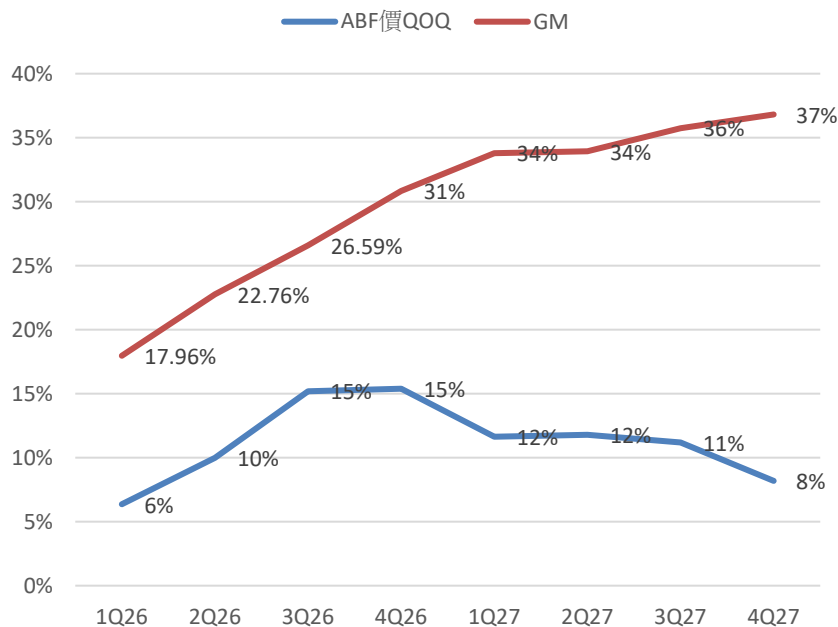


資料來源：元富投顧

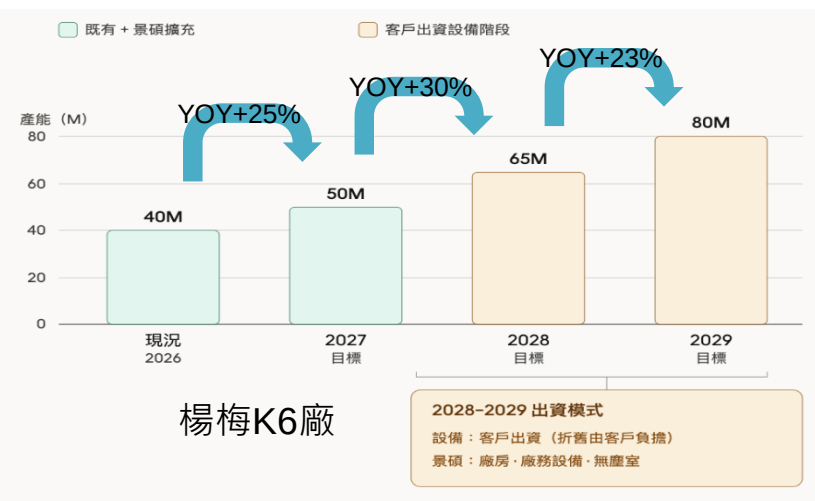


# 景碩(3189 TT,TP=NT\$1250)積極擴產搶攻市佔

| Sales(NT\$mn) | 2025   | %    | YoY | 2026(F) | %    | YoY | 2027(F) | %    | YoY |
|---------------|--------|------|-----|---------|------|-----|---------|------|-----|
| Substrate     | 32,312 | 82%  | 36% | 48,045  | 86%  | 49% | 77,967  | 90%  | 62% |
| ABF           | 17,912 | 46%  | 51% | 27,022  | 48%  | 51% | 49,949  | 58%  | 85% |
| BT            | 14,400 | 37%  | 21% | 21,023  | 38%  | 46% | 28,018  | 32%  | 33% |
| 隱形眼鏡          | 7,039  | 18%  | 3%  | 8,006   | 14%  | 14% | 8,806   | 10%  | 10% |
| 合計            | 39,351 | 100% | 29% | 56,051  | 100% | 42% | 86,773  | 100% | 55% |



- 景碩AI CPU載板2Q26開始放量；景碩Share 推估：2025 10% → 2026至少30-40% → 2027至少40-50%
- 景碩ABF擴產計畫：



# 景碩(3189 TT,TP=NT\$1250)積極擴產搶攻市佔

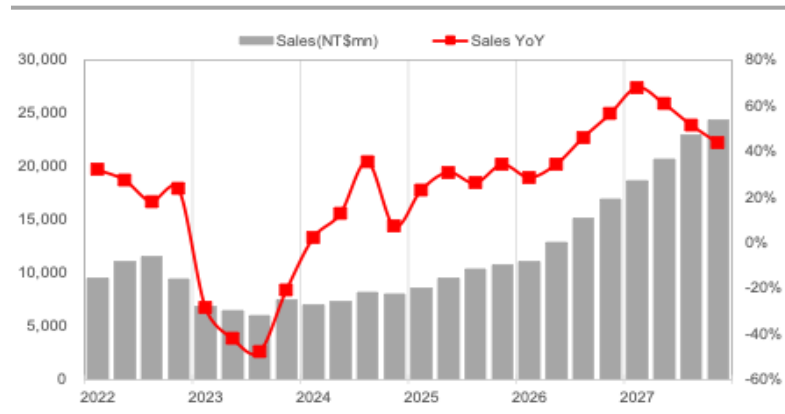
## Comprehensive income statement

NT\$m

| Comprehensive income statement |              |              |              |               | NT\$m |
|--------------------------------|--------------|--------------|--------------|---------------|-------|
| Year-end Dec. 31               | FY24         | FY25         | FY26 F       | FY27 F        |       |
|                                | IFRS         | IFRS         | IFRS         | IFRS          |       |
| Net sales                      | 30,535       | 39,351       | 55,678       | 86,363        |       |
| COGS                           | 21,867       | 31,037       | 40,237       | 56,739        |       |
| Gross profit                   | 8,668        | 8,314        | 15,442       | 29,624        |       |
| Operating expense              | 7,572        | 5,645        | 6,992        | 10,816        |       |
| <b>Operating profit</b>        | <b>1,095</b> | <b>2,670</b> | <b>8,450</b> | <b>18,808</b> |       |
| Total non-operate. Inc.        | 508          | 417          | 248          | 277           |       |
| Pre-tax profit                 | 1,603        | 3,087        | 8,698        | 19,086        |       |
| Total Net profit               | 1,331        | 2,717        | 7,836        | 17,177        |       |
| Minority                       | 0            | 0            | 0            | 0             |       |
| <b>Net Profit</b>              | <b>49</b>    | <b>1,596</b> | <b>6,532</b> | <b>15,750</b> |       |
| <b>EPS (NT\$)</b>              | <b>0.11</b>  | <b>3.51</b>  | <b>12.53</b> | <b>29.89</b>  |       |
| Y/Y %                          | FY24         | FY25         | FY26 F       | FY27 F        |       |
| Sales                          | 13.8         | 28.9         | 41.5         | 55.1          |       |
| Gross profit                   | 1,148.5      | -4.1         | 85.7         | 91.8          |       |
| Operating profit               | 387.3        | 143.7        | 216.5        | 122.6         |       |
| Pre-tax profit                 | 343.6        | 92.6         | 181.7        | 119.4         |       |
| Net profit                     | 虧轉盈          | 3,164.4      | 309.3        | 141.1         |       |
| EPS                            | -97.2        | 3,090.9      | 256.8        | 138.6         |       |
| Margins %                      | FY24         | FY25         | FY26 F       | FY27 F        |       |
| Gross                          | 28.4         | 21.1         | 27.7         | 34.3          |       |
| Operating                      | 3.6          | 6.8          | 15.2         | 21.8          |       |
| EBITDA                         | 23.6         | 25.9         | 0.0          | 0.0           |       |
| Pre-tax                        | 5.3          | 7.8          | 15.6         | 22.1          |       |
| <b>Net</b>                     | <b>0.2</b>   | <b>4.1</b>   | <b>11.7</b>  | <b>18.2</b>   |       |

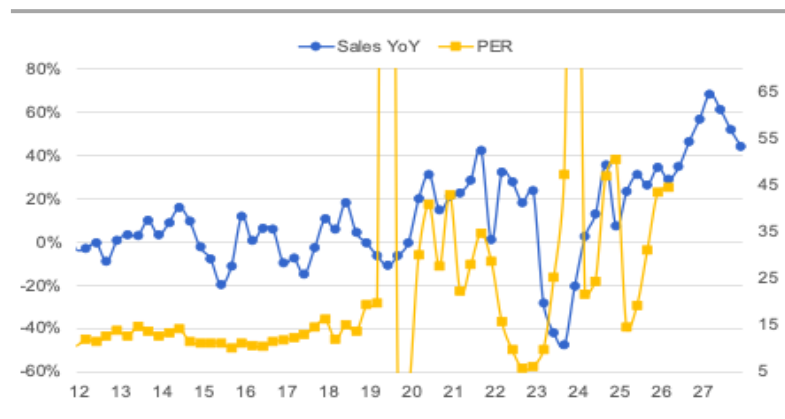
資料來源：元富投顧

## 營收與營收YOY



資料來源：元富投顧

## PER



資料來源：元富投顧

# 南電(8046 TT,TP=NT\$1700)現貨報價靈活調整

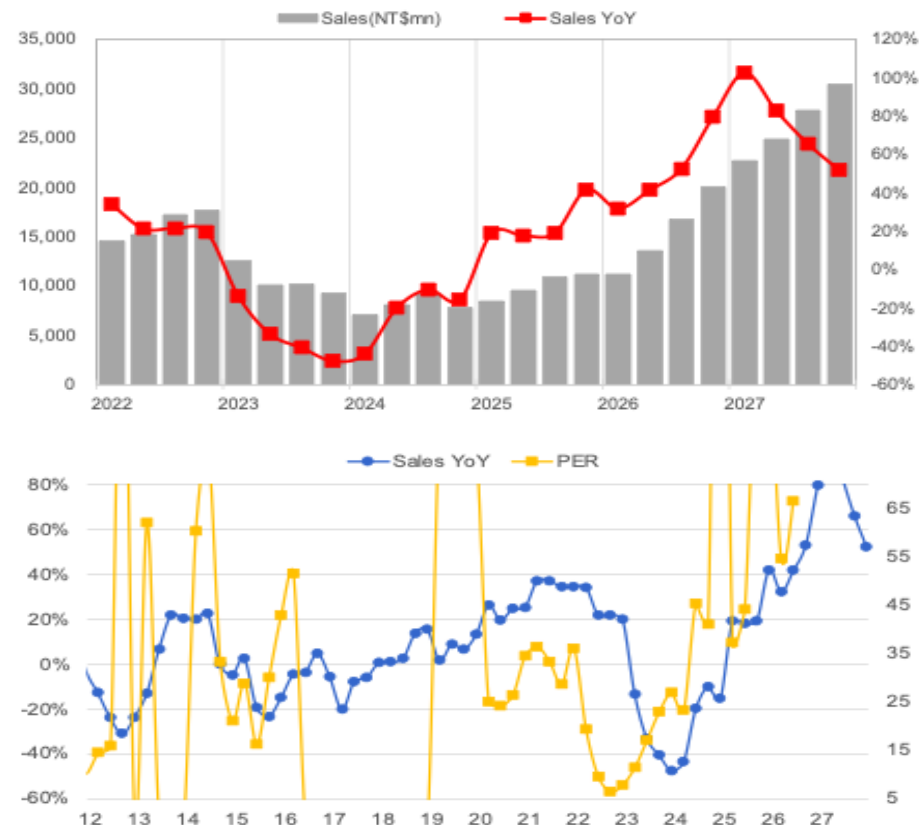
## Comprehensive income statement

NT\$m

| Year-end Dec. 31        | FY24           | FY25         | FY26 F        | FY27 F        |
|-------------------------|----------------|--------------|---------------|---------------|
|                         | IFRS           | IFRS         | IFRS          | IFRS          |
| Net sales               | 32,283         | 40,173       | 61,575        | 105,849       |
| COGS                    | 31,926         | 36,526       | 42,923        | 59,941        |
| Gross profit            | 358            | 3,647        | 18,652        | 45,908        |
| Operating expense       | 1,624          | 1,665        | 1,916         | 2,404         |
| <b>Operating profit</b> | <b>(1,267)</b> | <b>1,982</b> | <b>16,736</b> | <b>43,503</b> |
| Total non-operate. Inc. | 1,430          | 363          | 746           | 857           |
| Pre-tax profit          | 163            | 2,346        | 17,482        | 44,360        |
| Total Net profit        | 204            | 1,947        | 13,574        | 34,379        |
| Minority                | 0              | 0            | 0             | 0             |
| <b>Net Profit</b>       | <b>204</b>     | <b>1,947</b> | <b>13,574</b> | <b>34,379</b> |
| <b>EPS (NT\$)</b>       | <b>0.31</b>    | <b>3.01</b>  | <b>21.01</b>  | <b>53.21</b>  |
| Y/Y %                   | FY24           | FY25         | FY26 F        | FY27 F        |
| Sales                   | -23.6          | 24.4         | 53.3          | 71.9          |
| Gross profit            | -48.5          | 919.6        | 411.4         | 146.1         |
| Operating profit        | 盈轉虧            | 虧轉盈          | 744.2         | 159.9         |
| Pre-tax profit          | -54.9          | 1,338.4      | 645.3         | 153.8         |
| Net profit              | -36.0          | 855.6        | 597.3         | 153.3         |
| EPS                     | -92.1          | 871.0        | 598.1         | 153.2         |
| Margins %               | FY24           | FY25         | FY26 F        | FY27 F        |
| Gross                   | 1.1            | 9.1          | 30.3          | 43.4          |
| Operating               | -3.9           | 4.9          | 27.2          | 41.1          |
| EBITDA                  | 20.5           | 23.0         | 40.1          | 48.7          |
| Pre-tax                 | 0.5            | 5.8          | 28.4          | 41.9          |
| <b>Net</b>              | <b>0.6</b>     | <b>4.8</b>   | <b>22.0</b>   | <b>32.5</b>   |

資料來源：元富投顧

| Sales(NT\$m) | 2025   | %    | YoY  | 2026(F) | %    | YoY | 2027(F) | %    | YoY  |
|--------------|--------|------|------|---------|------|-----|---------|------|------|
| 電腦           | 6,829  | 17%  | 24%  | 10,468  | 17%  | 53% | 15,877  | 15%  | 52%  |
| 網路通訊         | 19,685 | 49%  | 33%  | 30,540  | 50%  | 55% | 53,983  | 51%  | 77%  |
| 消費性電子        | 4,821  | 12%  | 36%  | 6,837   | 11%  | 42% | 9,526   | 9%   | 39%  |
| 車用電子         | 2,410  | 6%   | -32% | 3,510   | 6%   | 46% | 5,292   | 5%   | 51%  |
| 人工智慧與高效運算    | 6,428  | 16%  | 33%  | 10,220  | 17%  | 59% | 21,170  | 20%  | 107% |
| 合計           | 40,173 | 100% | 24%  | 61,575  | 100% | 53% | 105,849 | 100% | 72%  |



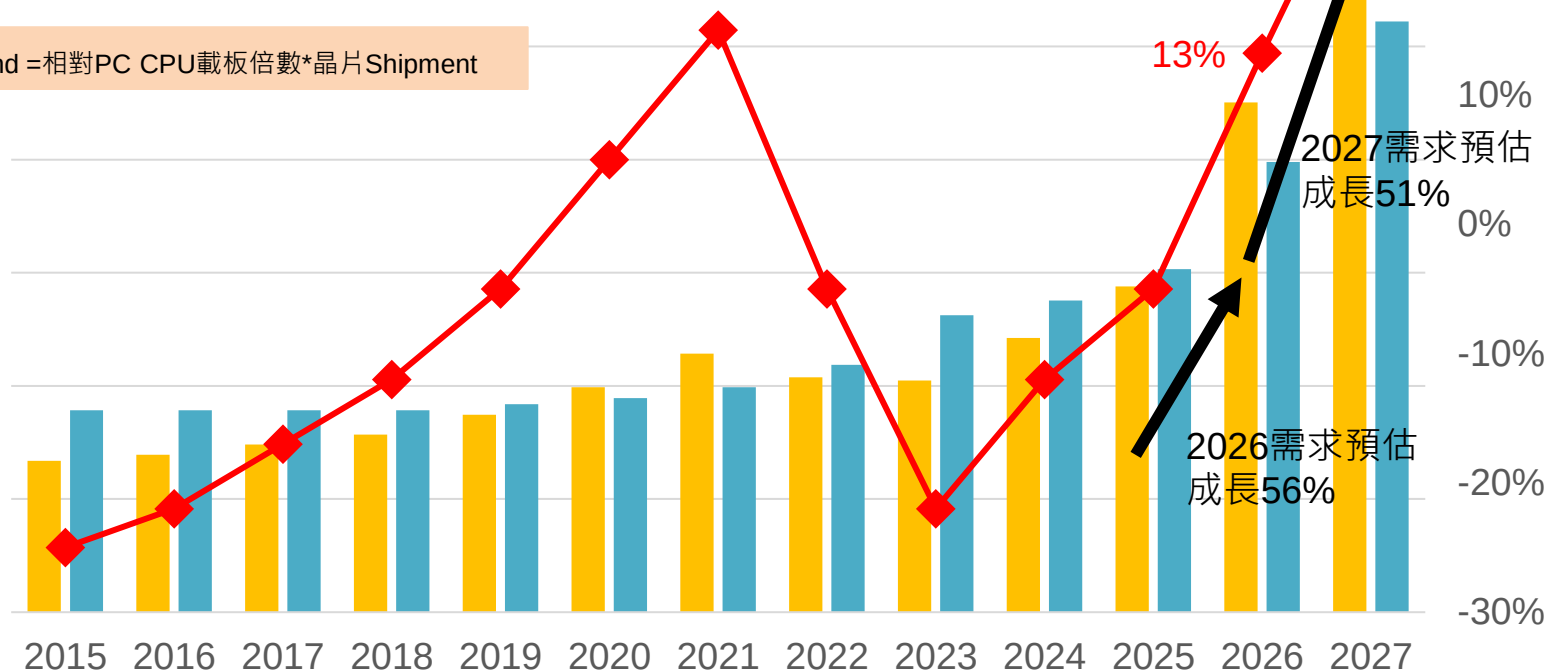
# 結論：預期2027年ABF載板供不應求幅度擴大

■ Demand ■ Supply ◆ 供需缺口(RHS)

2026-2027需求成長>載板廠產能擴充幅度；假設2026-2027全球載板廠平均擴產25%(上游材料、生產良率等皆為瓶頸，擴充幅度>有效產能)

|            | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 |
|------------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| Demand     | 134  | 139  | 148  | 157  | 175  | 199  | 229  | 208  | 205  | 242  | 288  | 450  | 682  |
| YOY        | -    | 4%   | 6%   | 6%   | 11%  | 14%  | 15%  | -9%  | -1%  | 18%  | 19%  | 56%  | 51%  |
| Supply     | 178  | 178  | 178  | 178  | 184  | 189  | 199  | 219  | 262  | 275  | 303  | 398  | 522  |
| 全球載板廠利用率推估 | 75%  | 78%  | 83%  | 88%  | 95%  | 105% | 115% | 95%  | 78%  | 88%  | 95%  | 105% | 105% |
| YOY        | 0%   | 0%   | 0%   | 3%   | 3%   | 5%   | 10%  | 20%  | 0%   | 5%   | 10%  | 25%  | 25%  |
| 供需缺口       | -25% | -22% | -17% | -12% | -5%  | 5%   | 15%  | -5%  | -22% | -12% | -5%  | 13%  | 31%  |

Demand = 相對PC CPU載板倍數\*晶片Shipment



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